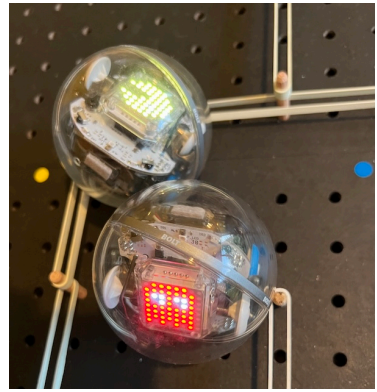




Robot Pac-Man



*Keynsham Music Festival is excited to announce a **Discovery Zone*** to this year's festival. The new space will give festivalgoers the chance to explore science, technology and research in a fun and interactive way alongside the music and community atmosphere the festival is known for.*

Code-Club present Robot Pac-Man. You've (probably) played Pac-Man the video game, but have you played it with robots? In this interactive exhibit, help Pac-Man escape the ghostly clutches of Blinky, Pinky, Inky, and Clyde.

This is the original Namco Pac-Man videogame from 1980. Can we recreate it with robots?

https://youtu.be/i_OjztdQ8iw



* You will need your parents' consent to help on Saturday July, 4.



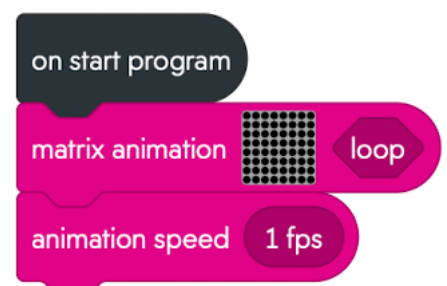
Ghosts

CHARACTER	/	NICKNAME
	- SHADOW	"BLINKY"
	- SPEEDY	"PINKY"
	- BASHFUL	"INKY"
	- POKEY	"CLYDE"

The Pac-Man videogame has four ghostly characters named "Blinky", "Pinky", "Inky" and "Clyde." We need at least one of each character, so see what others are doing, and do something different.

- 1) In Chrome, open <https://edu.sphero.com/code>
- 2) Connect your **Sphero BOLT** robot: click 
- 3) Select  then 
- 4) Your BOLT has a sticker with a code on it. Choose the matching BOLT from the list then 
- 5) Check your robot is showing the right image then 
- 6) Click **+ CREATE PROGRAM.** 
- 7) Name your program after the character you're creating. Leave **BLOCKS** selected. Select **SPHERO BOLT**, deselect **SPHERO BOLT+** then **CREATE.**

- 8) Add code for a **matrix animation** with *your ghost* design.



Click on the matrix grid to edit then  to add a new 8x8 matrix.

Click  to finish.

9) Animate it to make its eyes look left  then right .

Select your graphic then click  to edit it.

Click  to add a new frame.

To remove a frame, click , select a frame then .

10)  your code. Does it work?

When ghosts run away from Pac-Man, they turn blue and take fright.
Add another matrix design for this.

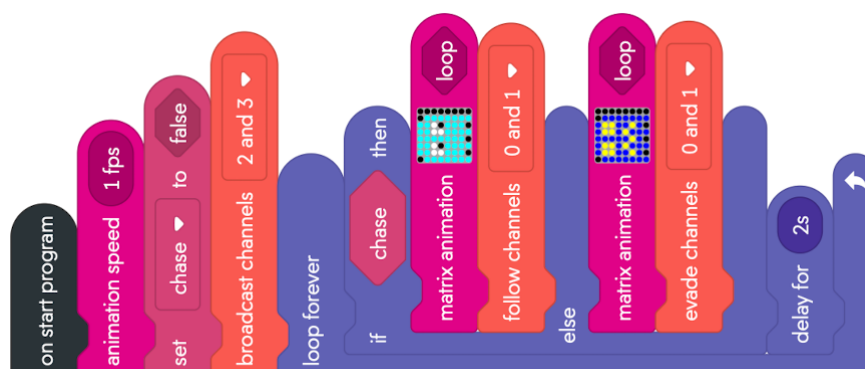


11) In **Variables**, create a new variable called **chase**.

This is **boolean** because it will only be *true* or *false*.

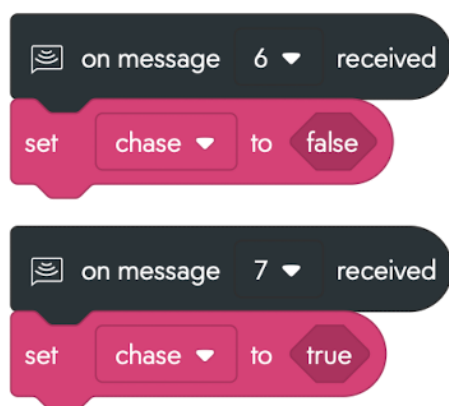
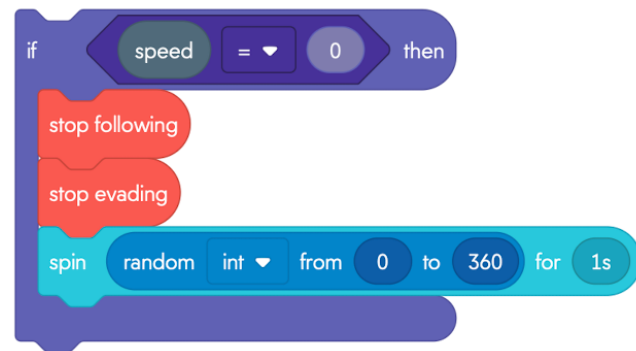
Sphero robots communicate using infra-red light invisible to us. Ghosts broadcast their position on channels 2, 3 (short & long range), and follow Pac-Man on channels 0, 1.

12) Replace the existing code with the following:



Sometimes the robot gets stuck on the maze walls.

13) Add the code (right) at the end of the loop (but still inside the loop) to detect when it has stopped moving and give it a random spin.

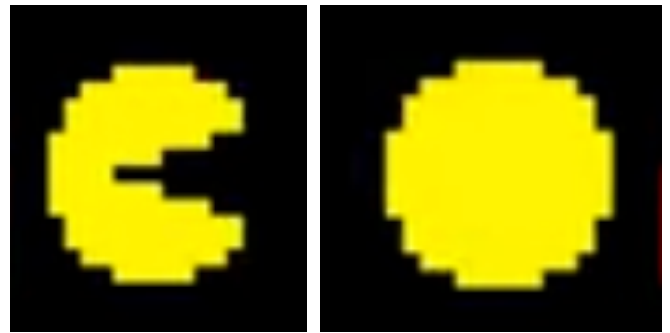


14) When Pac-Man receives an **energizer** it sends a message to reverse the **chase**. Add code (left) to receive this message and set chase to **true** or **false**.

To **Save** your code go back to the main page. Don't forget to add "Blinky", "Pinky", "Inky" or "Clyde" to the name of your program. Click on the 3 dots (ellipsis) next to your program and download to a USB drive.



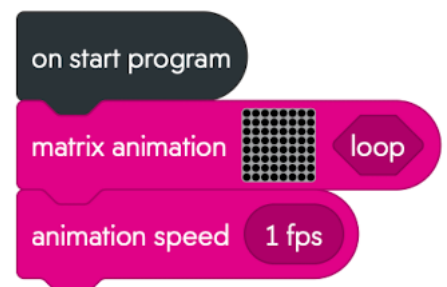
Pac-Man



*Pac-Man explores a maze, avoiding ghosts. An **energizer** temporarily reverses the chase where ghosts enter a frightened state and become vulnerable.*

- 1) In Chrome, open <https://edu.sphero.com/code>
- 2) Connect your **Sphero BOLT** robot: click 
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- 4) Your BOLT has a sticker with a code on it. Choose the matching BOLT from the list then 
- 5) Check your robot is showing the right image then 
- 6) Click **+ CREATE PROGRAM.** 
- 7) Name your program after the character you're creating. Leave **BLOCKS** selected. Select **SPHERO BOLT**, **deselect SPHERO BOLT+** then **CREATE.**

- 8) Add code for a **matrix animation** with *your Pac-Man* design. Animate it to make its mouth open and close.



Click on the matrix grid to edit then  to add a new 8x8 matrix.

Click  to finish.

Select your graphic then click  to edit it.

Click  to add a new frame.

To remove a frame, click , select a frame then .

9)  your code. Does it work?

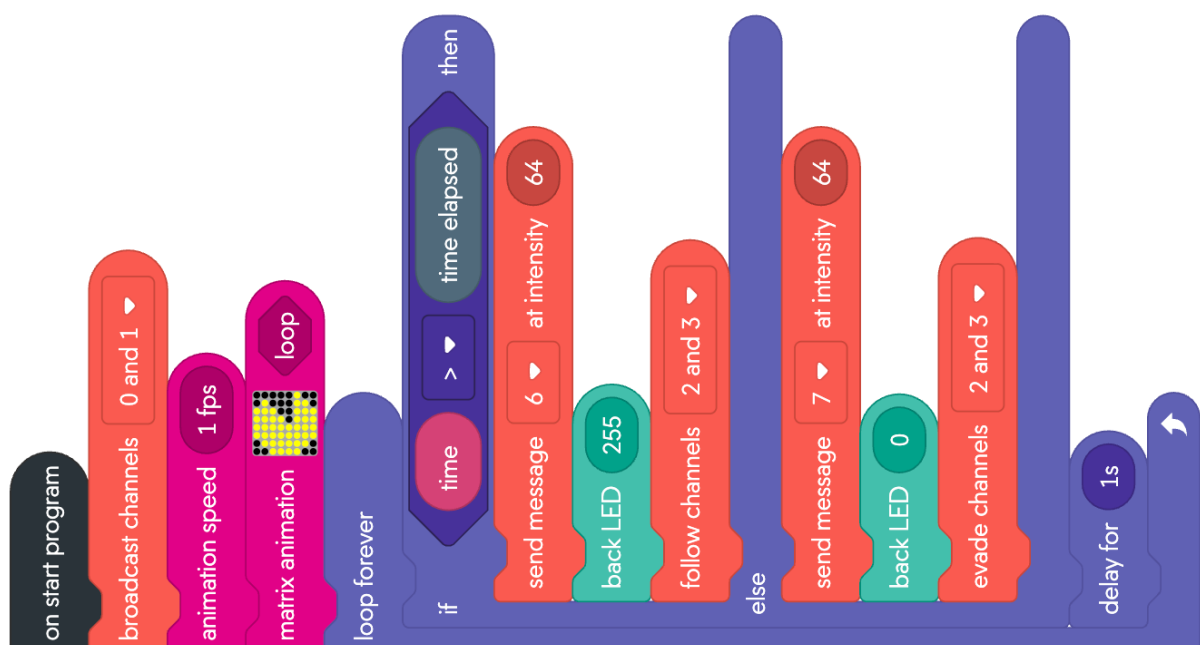
10) In **Variables**, create a new variable called **time**.

This is a **number** because it records time elapsed.

We'll use time for the **energizer** later.

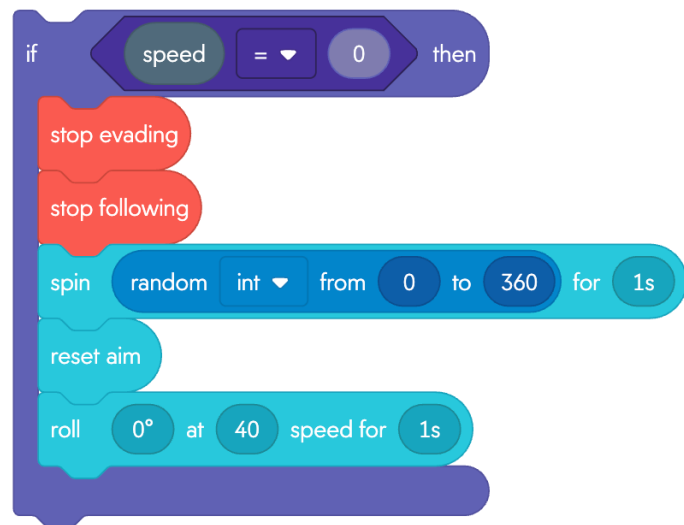
Sphero robots communicate using infra-red light invisible to us. Ghosts broadcast their position on channels 2, 3 (short & long range), and follow Pac-Man on channels 0, 1.

11) Replace the existing code with the following:



Sometimes Pac-Man gets stuck on the maze walls.

12) Add the code (right) at the end of the loop (but still inside the loop) to detect when it has stopped moving, give it a random spin and roll forwards.



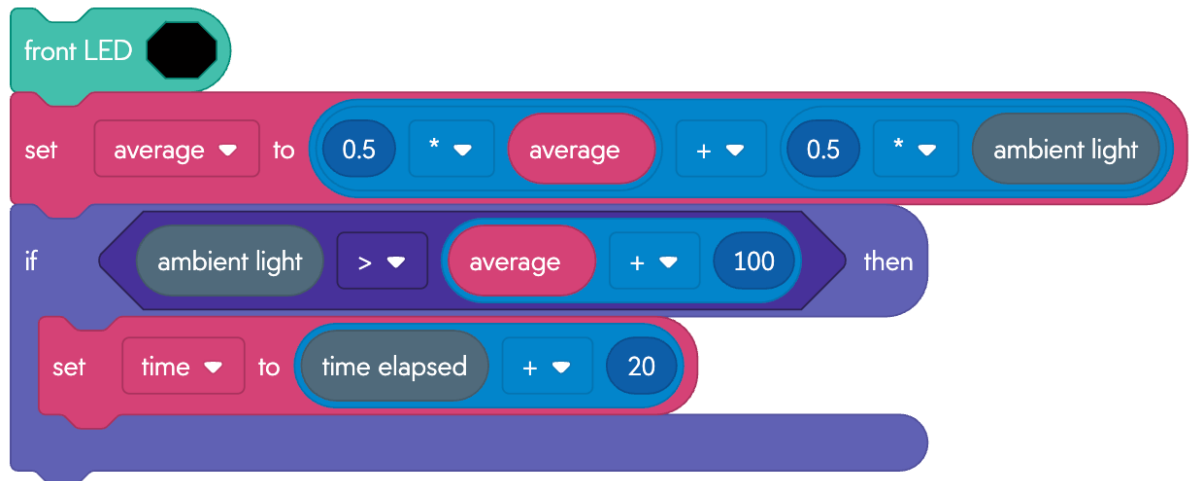
Energizer

In the videogame, the energizer is a large “Power Pill” that Pac-Man can eat to power-up. In the robot game we will use a torch to energize Pac-Man.

When we walk from bright sunshine into a house, we can’t see anything until our eyes adjust to the dark. The robot is the same. It has a light sensor, and we measure the average light level.

If the brightness suddenly increases, then assume that’s a flash of light from the torch. The light sensor is next to the front LED, so set this to a really dim red (red=1) to show people where to point the torch.

- 13) Add the code (below) at the end of the loop (still inside the loop) to take the average. If the brightness increases by 100 then set **time** to 20 seconds in the future, during which Pac-Man is **energized**.



To **Save** your code go back to the main page. Don't forget to add "Pac-Man" to the name of your program. Click on the 3 dots (ellipsis) next to your program and download to a USB drive.